

figures-referenced

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
library(ggsignif)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

gradient_it <- read_csv(here("data", "it_gradient_data.csv"))
```

```

metadata <- read_csv(here("data", "metadata.csv"))

site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

trail_creek_isco <- compiled_isotope |>
  filter(site %in% c("utc_isco", "tc_isco"))

trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

coal_creek <- compiled_isotope |>
  filter(location %in% c("cc"))

italian_creek <- compiled_isotope |>
  filter(location == "it")

trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
    delta_d = processed_delta_d,
    delta_d_std = processed_delta_d_stddev,
    delta_o = processed_delta_18o,
    delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

tc_isco <- trail_creek_isco |>
  filter(site == "tc_isco")

# Downstream Trail Creek
tc_metrics <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"), show_col_types = FALSE)
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,
    temp = temperature_c,
    water_level = water_level_m,

```

```

    bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
       date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
          avg_ap = mean(ap, na.rm = TRUE),
          avg_temp = mean(temp, na.rm = TRUE),
          avg_water_level = mean(water_level, na.rm = TRUE),
          avg_bp = mean(bp, na.rm = TRUE),
          total_readings = n(), # the amount of data it used to average
          .groups = "drop") |>
mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_metrics <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types = FALSE) |> # 1
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
       date = as_date(date_time_parsed)) |> # making a date column
group_by(date) |>
summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
          avg_ap = mean(ap, na.rm = TRUE),
          avg_temp = mean(temp, na.rm = TRUE),
          avg_water_level = mean(water_level, na.rm = TRUE),
          avg_bp = mean(bp, na.rm = TRUE),
          total_readings = n(), # the amount of data it used to average
          .groups = "drop") |>
mutate(site = "utc") # creating a new column of the site

tc_utc_metrics <- merge(tc_metrics, utc_metrics, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_metrics,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)

```

```

discharge_tc_isco <- read_csv(here("data", "discharge_tc_isco.csv"))

discharge_utc_isco <- read_csv(here("data", "discharge_utc_isco.csv"))

discharge_isco <- read_csv(here("data", "isco_discharge_lc.csv"))

```

Figures

Figure 1: Time-Series of Line-Conditioned Excess (lc-excess) of ISCO's

```

ggplot(data = trail_creek_isco,
  aes(x = date,
    y = lc_excess,
    color = site,
    group = site)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +

  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\n", \n"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
    label.y = "top", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.5) +
  scale_color_manual(
    values = c("tc_isco" = "dodgerblue", "utc_isco" = "orangered2"),
    labels = c("tc_isco" = "Trail Creek", "utc_isco" = "Upper Trail Creek")
  ) +

  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",

```

```

y = "lc-excess (\u2030)",
color = "Site") +
ylim(-8,5) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")

```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment

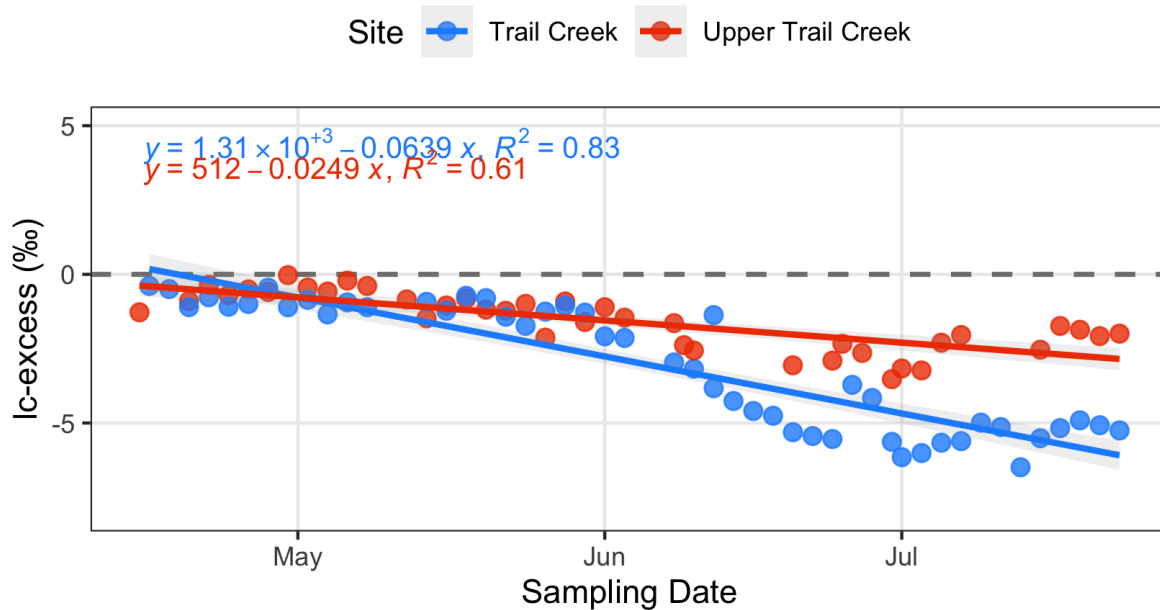


Figure 2: Sampling Site and lc-excess (Trail Creek)

```

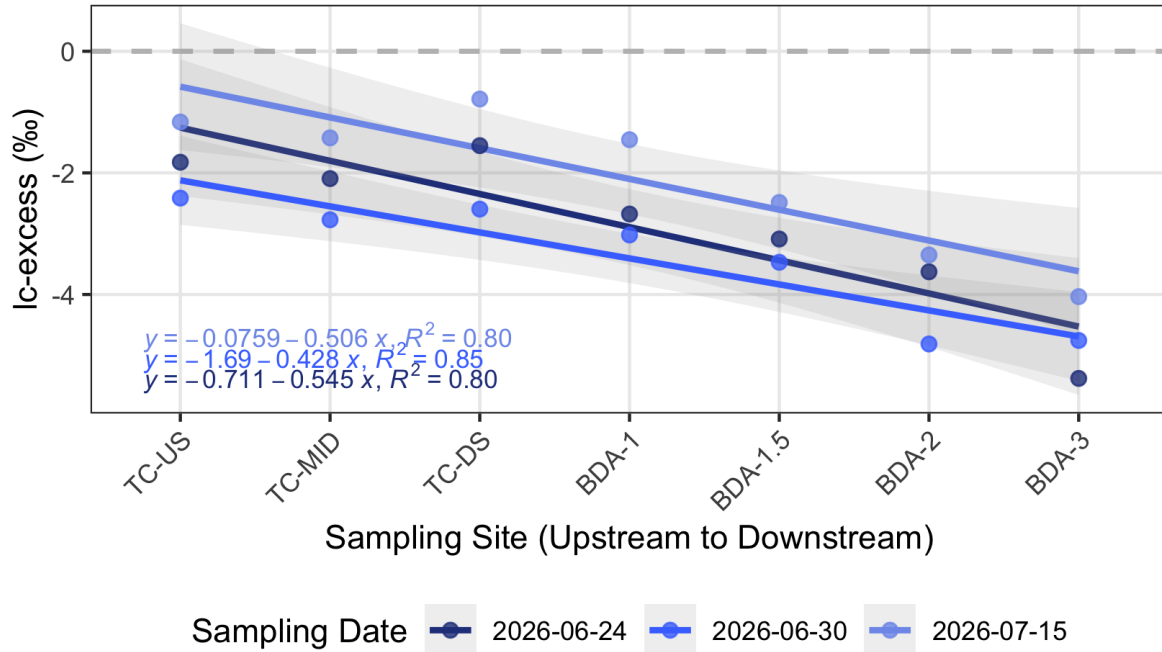
plot2_1 <- ggplot(data = trail_creek, aes(x = site, y = lc_excess, color = factor(date), group = date)) +
  # reference line at 0
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  # adds a linear regression line for each date group
  geom_smooth(method = "lm", se = TRUE, linewidth = 1, alpha = 0.15) +
  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\n", \n")),
    formula = y ~ x,
    parse = TRUE,

```

```

label.x = "left", # positions the labels horizontally
label.y = "bottom", # positions the labels vertically
vstep = 0.05, # spacing between the text lines so they do not overlap
size = 3.0) +
# points for each site's value
geom_point(size = 2.0, alpha = 0.8) +
labs(
  title = "Trail Creek",
  x = "Sampling Site (Upstream to Downstream)",
  y = "lc-excess (\u2030)",
  color = "Sampling Date"
) +
scale_color_manual(values = c("2026-06-24" = "royalblue4", "2026-06-30" = "royalblue1", "2026-07-07" = "royalblue1", "2026-07-14" = "royalblue1", "2026-07-21" = "royalblue1", "2026-07-28" = "royalblue1", "2026-08-04" = "royalblue1", "2026-08-11" = "royalblue1", "2026-08-18" = "royalblue1", "2026-08-25" = "royalblue1", "2026-09-01" = "royalblue1", "2026-09-08" = "royalblue1", "2026-09-15" = "royalblue1", "2026-09-22" = "royalblue1", "2026-09-29" = "royalblue1", "2026-10-06" = "royalblue1", "2026-10-13" = "royalblue1", "2026-10-20" = "royalblue1", "2026-10-27" = "royalblue1", "2026-11-03" = "royalblue1", "2026-11-10" = "royalblue1", "2026-11-17" = "royalblue1", "2026-11-24" = "royalblue1", "2026-12-01" = "royalblue1", "2026-12-08" = "royalblue1", "2026-12-15" = "royalblue1", "2026-12-22" = "royalblue1", "2026-12-29" = "royalblue1", "2027-01-05" = "royalblue1", "2027-01-12" = "royalblue1", "2027-01-19" = "royalblue1", "2027-01-26" = "royalblue1", "2027-02-02" = "royalblue1", "2027-02-09" = "royalblue1", "2027-02-16" = "royalblue1", "2027-02-23" = "royalblue1", "2027-03-01" = "royalblue1", "2027-03-08" = "royalblue1", "2027-03-15" = "royalblue1", "2027-03-22" = "royalblue1", "2027-03-29" = "royalblue1", "2027-04-05" = "royalblue1", "2027-04-12" = "royalblue1", "2027-04-19" = "royalblue1", "2027-04-26" = "royalblue1", "2027-05-03" = "royalblue1", "2027-05-10" = "royalblue1", "2027-05-17" = "royalblue1", "2027-05-24" = "royalblue1", "2027-05-31" = "royalblue1", "2027-06-07" = "royalblue1", "2027-06-14" = "royalblue1", "2027-06-21" = "royalblue1", "2027-06-28" = "royalblue1", "2027-07-05" = "royalblue1", "2027-07-12" = "royalblue1", "2027-07-19" = "royalblue1", "2027-07-26" = "royalblue1", "2027-08-02" = "royalblue1", "2027-08-09" = "royalblue1", "2027-08-16" = "royalblue1", "2027-08-23" = "royalblue1", "2027-08-30" = "royalblue1", "2027-09-06" = "royalblue1", "2027-09-13" = "royalblue1", "2027-09-20" = "royalblue1", "2027-09-27" = "royalblue1", "2027-10-04" = "royalblue1", "2027-10-11" = "royalblue1", "2027-10-18" = "royalblue1", "2027-10-25" = "royalblue1", "2027-11-01" = "royalblue1", "2027-11-08" = "royalblue1", "2027-11-15" = "royalblue1", "2027-11-22" = "royalblue1", "2027-11-29" = "royalblue1", "2027-12-06" = "royalblue1", "2027-12-13" = "royalblue1", "2027-12-20" = "royalblue1", "2027-12-27" = "royalblue1", "2028-01-03" = "royalblue1", "2028-01-10" = "royalblue1", "2028-01-17" = "royalblue1", "2028-01-24" = "royalblue1", "2028-01-31" = "royalblue1", "2028-02-07" = "royalblue1", "2028-02-14" = "royalblue1", "2028-02-21" = "royalblue1", "2028-02-28" = "royalblue1", "2028-03-06" = "royalblue1", "2028-03-13" = "royalblue1", "2028-03-20" = "royalblue1", "2028-03-27" = "royalblue1", "2028-04-03" = "royalblue1", "2028-04-10" = "royalblue1", "2028-04-17" = "royalblue1", "2028-04-24" = "royalblue1", "2028-05-01" = "royalblue1", "2028-05-08" = "royalblue1", "2028-05-15" = "royalblue1", "2028-05-22" = "royalblue1", "2028-05-29" = "royalblue1", "2028-06-05" = "royalblue1", "2028-06-12" = "royalblue1", "2028-06-19" = "royalblue1", "2028-06-26" = "royalblue1", "2028-07-03" = "royalblue1", "2028-07-10" = "royalblue1", "2028-07-17" = "royalblue1", "2028-07-24" = "royalblue1", "2028-07-31" = "royalblue1", "2028-08-07" = "royalblue1", "2028-08-14" = "royalblue1", "2028-08-21" = "royalblue1", "2028-08-28" = "royalblue1", "2028-09-04" = "royalblue1", "2028-09-11" = "royalblue1", "2028-09-18" = "royalblue1", "2028-09-25" = "royalblue1", "2028-10-02" = "royalblue1", "2028-10-09" = "royalblue1", "2028-10-16" = "royalblue1", "2028-10-23" = "royalblue1", "2028-10-30" = "royalblue1", "2028-11-06" = "royalblue1", "2028-11-13" = "royalblue1", "2028-11-20" = "royalblue1", "2028-11-27" = "royalblue1", "2028-12-04" = "royalblue1", "2028-12-11" = "royalblue1", "2028-12-18" = "royalblue1", "2028-12-25" = "royalblue1", "2029-01-01" = "royalblue1", "2029-01-08" = "royalblue1", "2029-01-15" = "royalblue1", "2029-01-22" = "royalblue1", "2029-01-29" = "royalblue1", "2029-02-05" = "royalblue1", "2029-02-12" = "royalblue1", "2029-02-19" = "royalblue1", "2029-02-26" = "royalblue1", "2029-03-05" = "royalblue1", "2029-03-12" = "royalblue1", "2029-03-19" = "royalblue1", "2029-03-26" = "royalblue1", "2029-04-02" = "royalblue1", "2029-04-09" = "royalblue1", "2029-04-16" = "royalblue1", "2029-04-23" = "royalblue1", "2029-04-30" = "royalblue1", "2029-05-07" = "royalblue1", "2029-05-14" = "royalblue1", "2029-05-21" = "royalblue1", "2029-05-28" = "royalblue1", "2029-06-04" = "royalblue1", "2029-06-11" = "royalblue1", "2029-06-18" = "royalblue1", "2029-06-25" = "royalblue1", "2029-07-02" = "royalblue1", "2029-07-09" = "royalblue1", "2029-07-16" = "royalblue1", "2029-07-23" = "royalblue1", "2029-07-30" = "royalblue1", "2029-08-06" = "royalblue1", "2029-08-13" = "royalblue1", "2029-08-20" = "royalblue1", "2029-08-27" = "royalblue1", "2029-09-03" = "royalblue1", "2029-09-10" = "royalblue1", "2029-09-17" = "royalblue1", "2029-09-24" = "royalblue1", "2029-10-01" = "royalblue1", "2029-10-08" = "royalblue1", "2029-10-15" = "royalblue1", "2029-10-22" = "royalblue1", "2029-10-29" = "royalblue1", "2029-11-05" = "royalblue1", "2029-11-12" = "royalblue1", "2029-11-19" = "royalblue1", "2029-11-26" = "royalblue1", "2029-12-03" = "royalblue1", "2029-12-10" = "royalblue1", "2029-12-17" = "royalblue1", "2029-12-24" = "royalblue1", "2030-01-01" = "royalblue1", "2030-01-08" = "royalblue1", "2030-01-15" = "royalblue1", "2030-01-22" = "royalblue1", "2030-01-29" = "royalblue1", "2030-02-05" = "royalblue1", "2030-02-12" = "royalblue1", "2030-02-19" = "royalblue1", "2030-02-26" = "royalblue1", "2030-03-05" = "royalblue1", "2030-03-12" = "royalblue1", "2030-03-19" = "royalblue1", "2030-03-26" = "royalblue1", "2030-04-02" = "royalblue1", "2030-04-09" = "royalblue1", "2030-04-16" = "royalblue1", "2030-04-23" = "royalblue1", "2030-04-30" = "royalblue1", "2030-05-07" = "royalblue1", "2030-05-14" = "royalblue1", "2030-05-21" = "royalblue1", "2030-05-28" = "royalblue1", "2030-06-04" = "royalblue1", "2030-06-11" = "royalblue1", "203
```

Trail Creek

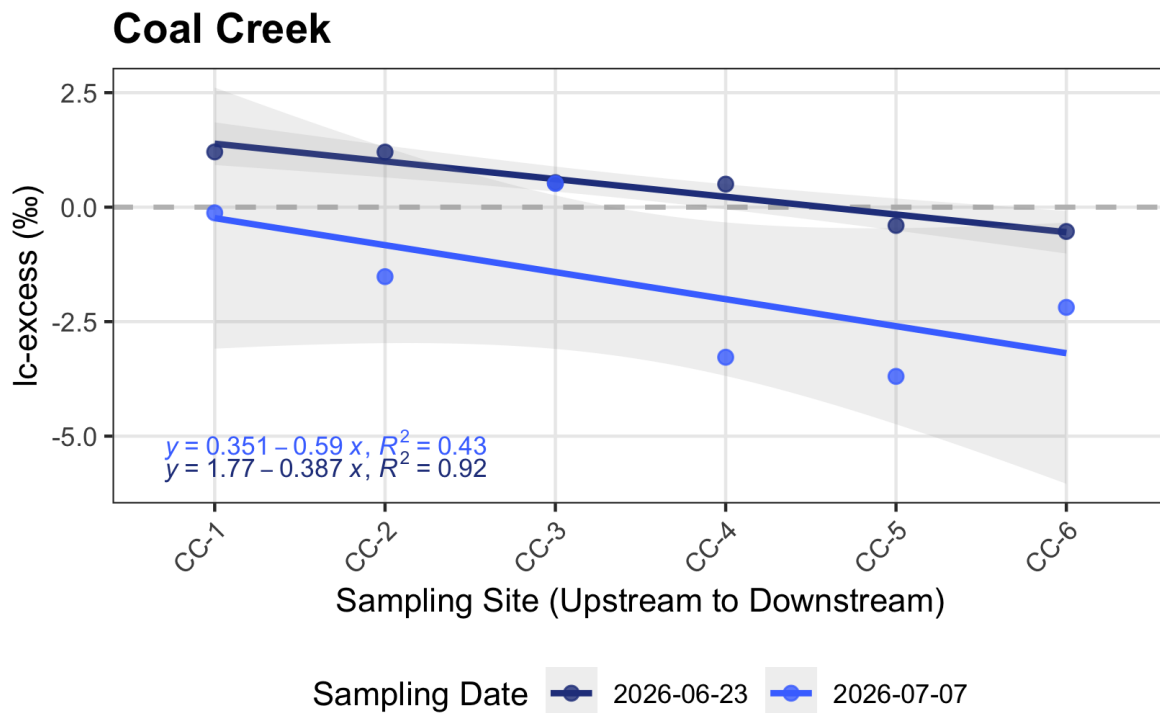


```
plot2_2 <- ggplot(data = coal_creek, aes(x = site, y = lc_excess, color = factor(date), group = date)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, linewidth = 1, alpha = 0.15) +
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\n", \n")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left",
    label.y = "bottom",
    vstep = 0.05,
    size = 3.0) +
  geom_point(size = 2.0, alpha = 0.8) +
  scale_color_manual(values = c("2026-06-23" = "royalblue4", "2026-07-07" = "royalblue1")) +
  scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  labs(
    title = "Coal Creek",
    x = "Sampling Site (Upstream to Downstream)",
    y = "lc-excess (\u2030)",
    color = "Sampling Date"
  ) +
```

```

theme_bw() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom"
)
plot2_2

```



```

plot2_3 <- ggplot(data = italian_creek, aes(x = site, y = lc_excess, color = factor(date), group = date)) +
  # reference line at 0
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  # adds a linear regression line for each date group
  geom_smooth(method = "lm", se = TRUE, linewidth = 1, alpha = 0.15) +
  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\n", \n")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
  )

```

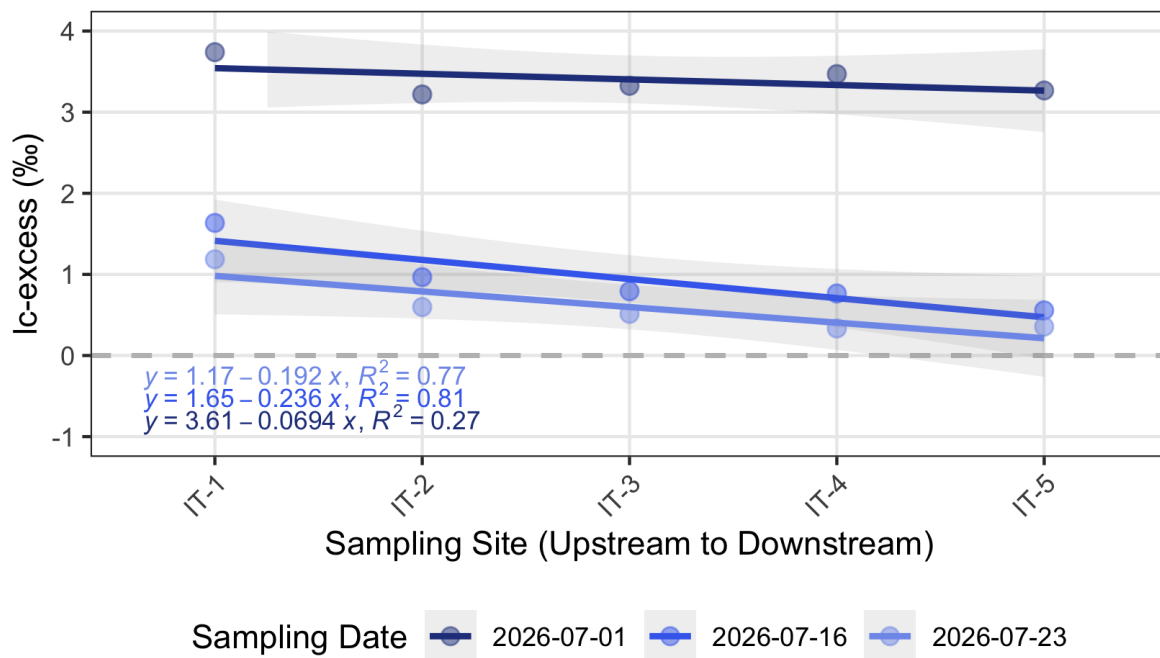


```

    label.y = "bottom", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.0) +
  ylim(-1,4) +
  # points for each site's value
  geom_point(size = 2.5, alpha = 0.5) +
  scale_color_manual(values = c("2026-07-01" = "royalblue4", "2026-07-16" = "royalblue2", "2026-07-23" = "royalblue1"),
  scale_x_discrete(labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3"),
  labs(
    title = "Italian Creek",
    x = "Sampling Site (Upstream to Downstream)",
    y = "lc-excess (\u2030)",
    color = "Sampling Date"
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
plot2_3

```

Italian Creek



```
figure_2 <- plot2_1 + plot2_2 + plot2_3
```

```
figure_2
```

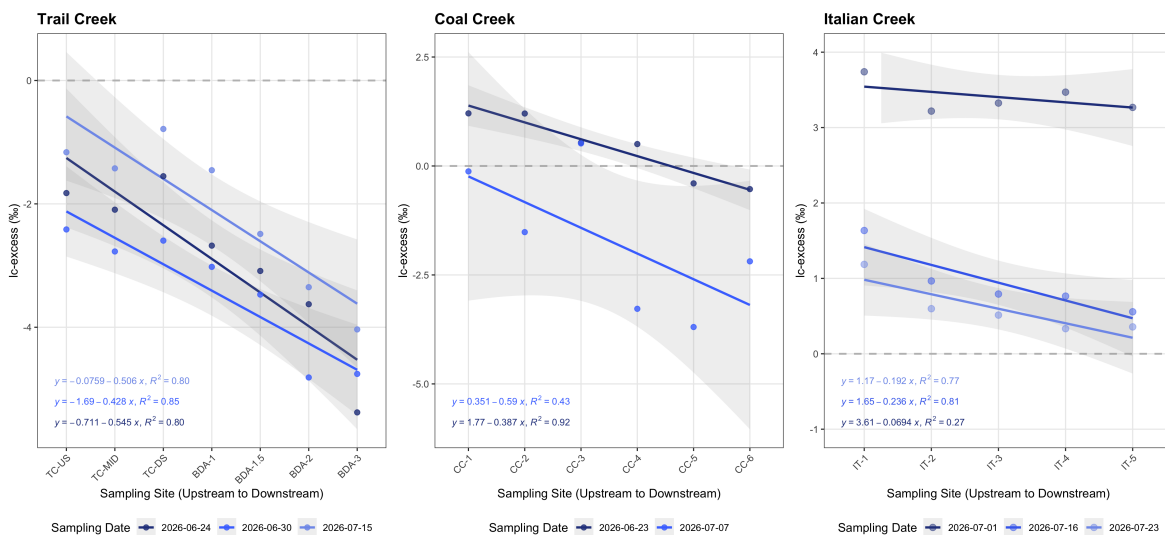


Figure 3:

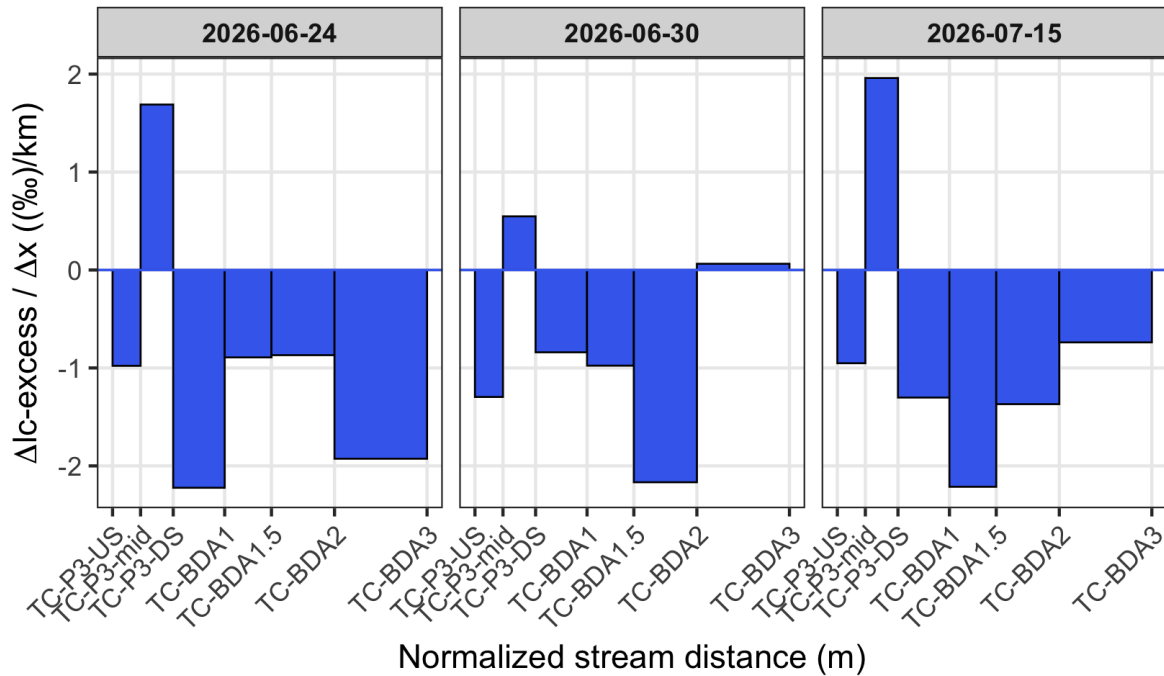
```
# Spatial variations in  $\Delta$ lc-excess/ $\Delta$ x across sampling dates
# Buildings a look up table of stream distances Trail Creek
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_tc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`) |>
  pivot_longer(cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`),
               names_to = "date",
               values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                          date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
plot3_1 <- ggplot(gradient_tc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))
```

plot3_1

Trail Creek



```
# Building a look up table of stream distances Coal Creek
distances_cc <- tibble(
  site = c("CC1", "CC2", "CC4", "CC5", "CC6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)) |>
mutate(xmax = dist_m,
       xmin = lag(dist_m, default = 0))

# Creating a filtered data frame for lc-excess
gradient_cc_clean <- gradient_cc |>
select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`) |>
pivot_longer(cols = c(`Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`),
             names_to = "date",
             values_to = "gradient") |>
mutate(date = case_when(
  date == "Δlc-excess/Δx_6-23" ~ "2026-06-23",
  date == "Δlc-excess/Δx_7-7" ~ "2026-07-07"),
       gradient = as.numeric(gradient)) |>
left_join(distances_cc |> select(site, xmin, xmax),
```

```

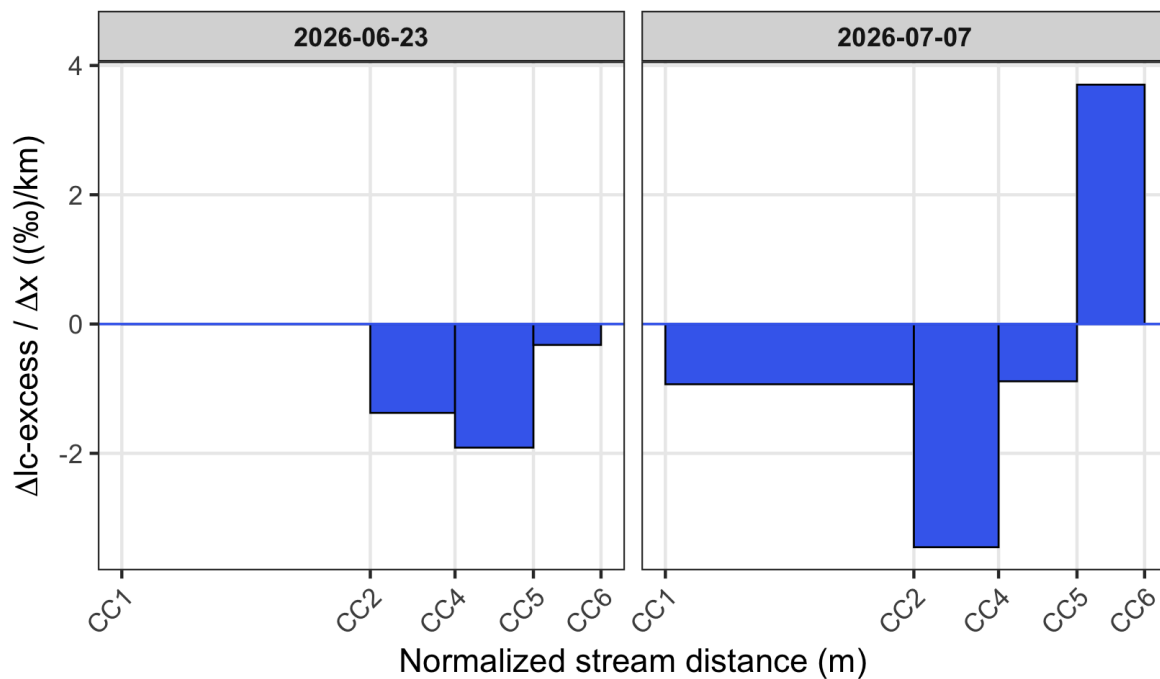
      by = "site") |>
mutate(site = factor(site, levels = distances_cc$site)) |>
filter(!is.na(gradient))

# Gradient plot
plot3_2 <- ggplot(gradient_cc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_cc$dist_m,
    labels = distances_cc$site) +
  labs(
    title = "Coal Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1))

plot3_2

```

Coal Creek



```
# Building a look up table of stream distances Coal Creek
distances_it <- tibble(
  site = c("IT1", "IT2", "IT3", "IT4", "IT5"),
  dist_m = c(0, 793.37, 1665.83, 2385.13, 2963.62)
) |>
mutate(
  xmax = dist_m,
  xmin = lag(dist_m, default = 0)
)

# Creating a filtered data frame for lc-excess
gradient_it_clean <- gradient_it |>
select(site, normalized_stream_distance_m, `d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`)
pivot_longer(
  cols = c(`d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`),
  names_to = "date",
  values_to = "gradient"
) |>
mutate(
  date = case_when(
    date == "d_lc/d_x_7_1" ~ "2026-07-01",

```

```

      date == "d_lc/d_x_7_16" ~ "2026-07-16",
      date == "d_lc/d_x_7_23" ~ "2026-07-23"
    ),
    gradient = as.numeric(gradient)
  ) |>
left_join(distances_it |> select(site, xmin, xmax),
          by = "site") |>
mutate(site = factor(site, levels = distances_it$site)) |> # FIXED: Uses distances_it inst
filter(!is.na(gradient))

# Gradient plot
plot3_3 <- ggplot(gradient_it_clean) +
  geom_rect(
    aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
    fill = "royalblue2", color = "black", linewidth = 0.3
  ) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_it$dist_m,
    labels = distances_it$site
  ) +
  labs(
    title = "Italian Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))
  ) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

plot3_3

```

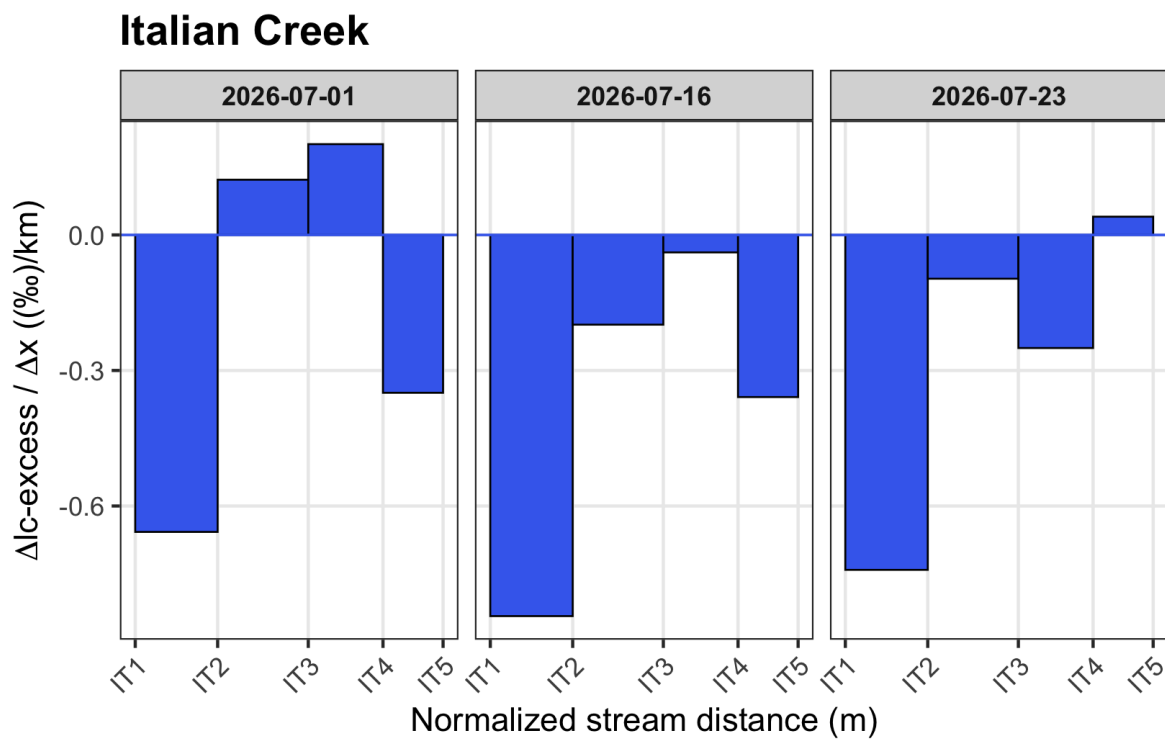


Figure 3

```
figure_3 <- plot3_1 + plot3_2 + plot3_3
figure_3 # print the graph
```

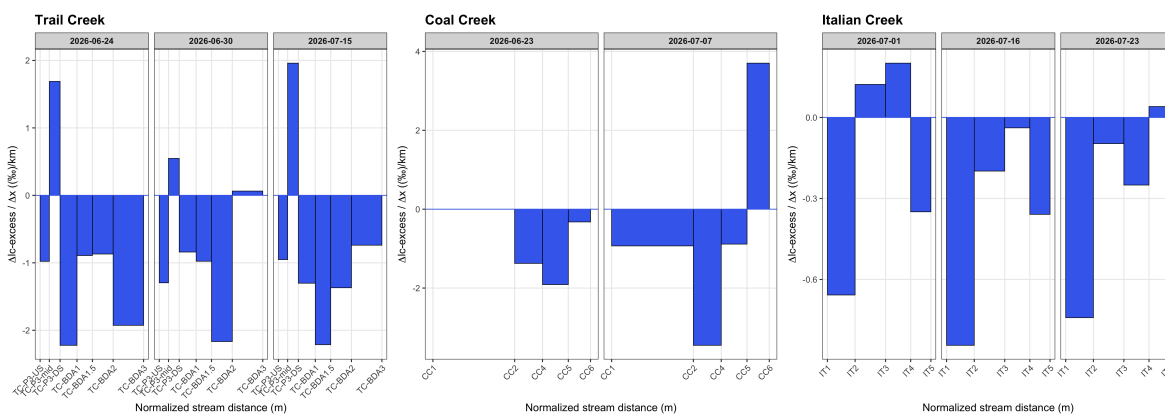
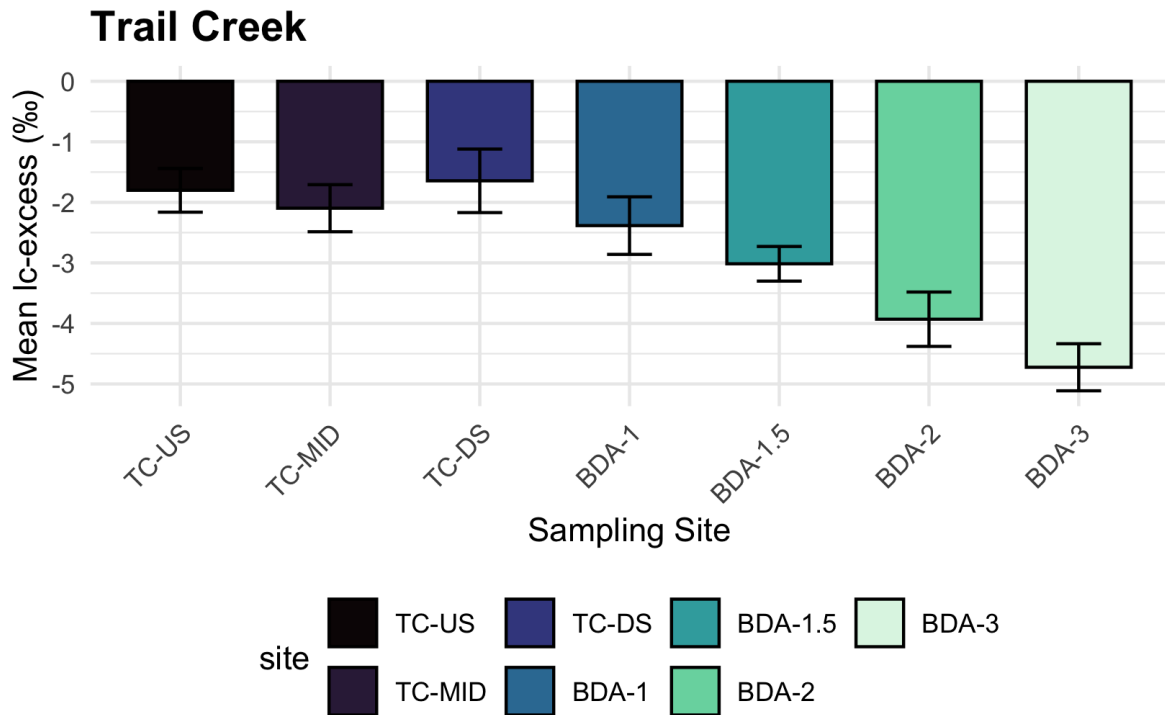


Figure 4

```
# averaging the lc-excess and including standard error
delta_summary <- trail_creek |>
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (coal creek)
plot4_1 <- ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3,
    color = "black") +
  scale_x_discrete(labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5",
    "tc_2" = "BDA-2", "tc_3" = "BDA-3",
    "tc_ds" = "TC-DS", "tc_mid" = "TC-MID",
    "tc_us" = "TC-US")) +
  scale_fill_viridis_d(option = "G", labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5", "tc_2" =
    "tc_3" = "BDA-3", "tc_ds" = "TC-DS", "tc_mid" = "TC-MID", "tc_us" = "TC-US")) +
  theme_minimal() +
  labs(title = "Trail Creek",
    x = "Sampling Site",
    y = "Mean lc-excess (\u2030)") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom")

plot4_1
```

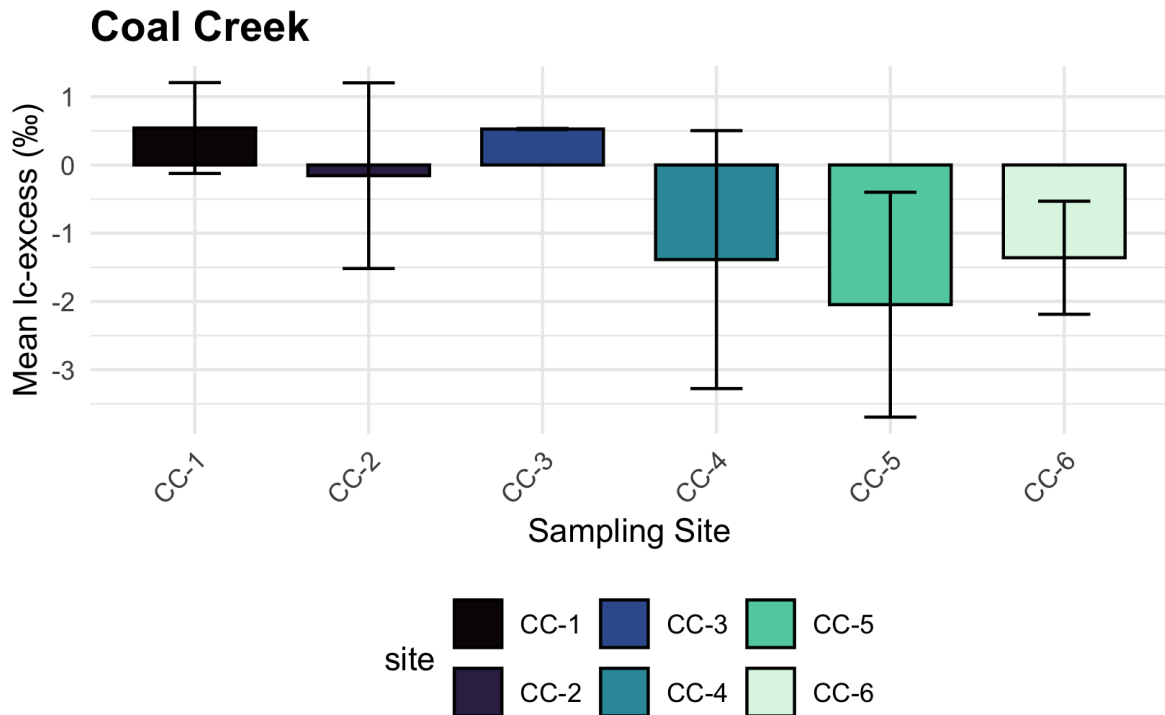


```
# averaging the lc-excess and including standard error
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (coal creek)
plot4_2 <- ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
    width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G", labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4" = "CC-4")) +
  theme_minimal() +
  labs(title = "Coal Creek",
    x = "Sampling Site",
    y = "Mean lc-excess (‰)") +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 45, hjust = 1),
```

```
legend.position = "bottom")
```

plot4_2



```
# averaging the lc-excess and including standard error
```

```
delta_summary_it <- italian_creek |>
```

```
  group_by(site) |>
```

```
  summarise(
```

```
    mean_lc = mean(lc_excess, na.rm = TRUE),
```

```
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
```

```
    .groups = "drop")
```

```
# creating a plot to show the mean lc-excess of each site (italian creek)
```

```
plot4_3 <- ggplot(delta_summary_it, aes(x = site, y = mean_lc, fill = site)) +
```

```
  geom_col(color = "black", width = 0.7) +
```

```
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
```

```
    width = 0.3, color = "black") +
```

```
  scale_fill_viridis_d(option = "G", labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3")) +
```

```
  scale_x_discrete(labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3")) +
```

```
  theme_minimal() +
```

```
labs(title = "Italian Creek",
     x = "Sampling Site",
     y = "Mean lc-excess (‰)") +
theme(plot.title = element_text(face = "bold", size = 14),
     axis.text.x = element_text(angle = 45, hjust = 1),
     legend.position = "bottom")
```

plot4_3

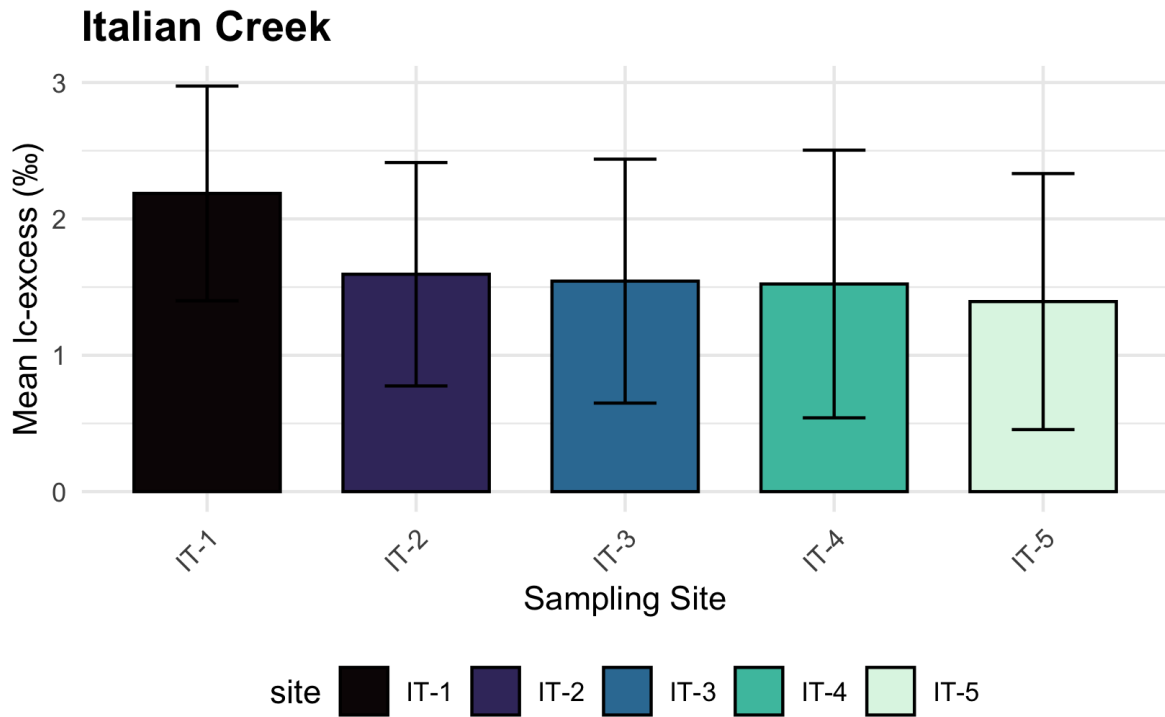


Figure 4:

```
figure_4 <- plot4_1 + plot4_2 + plot4_3
figure_4 # print the graph
```

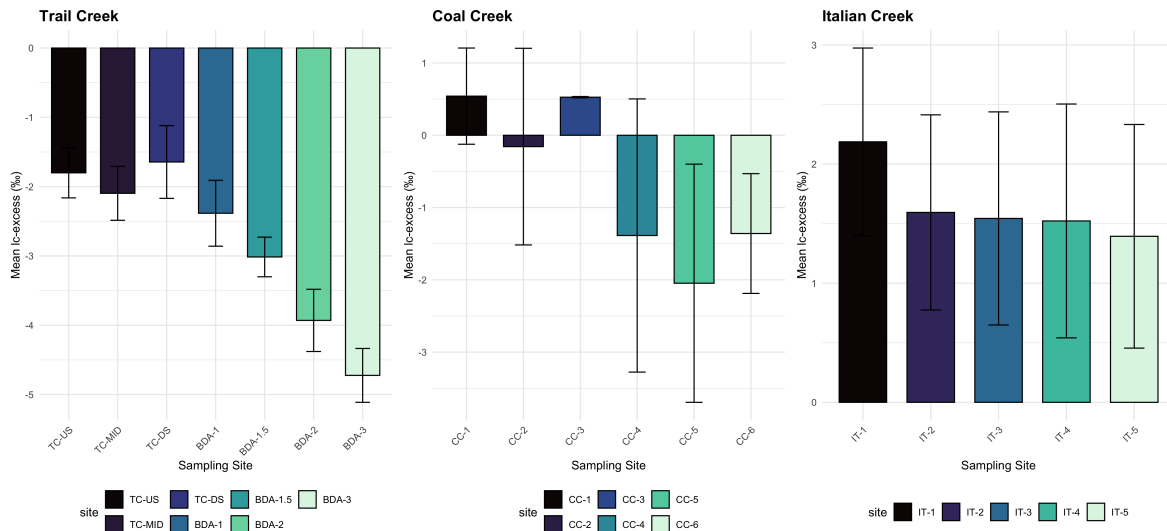
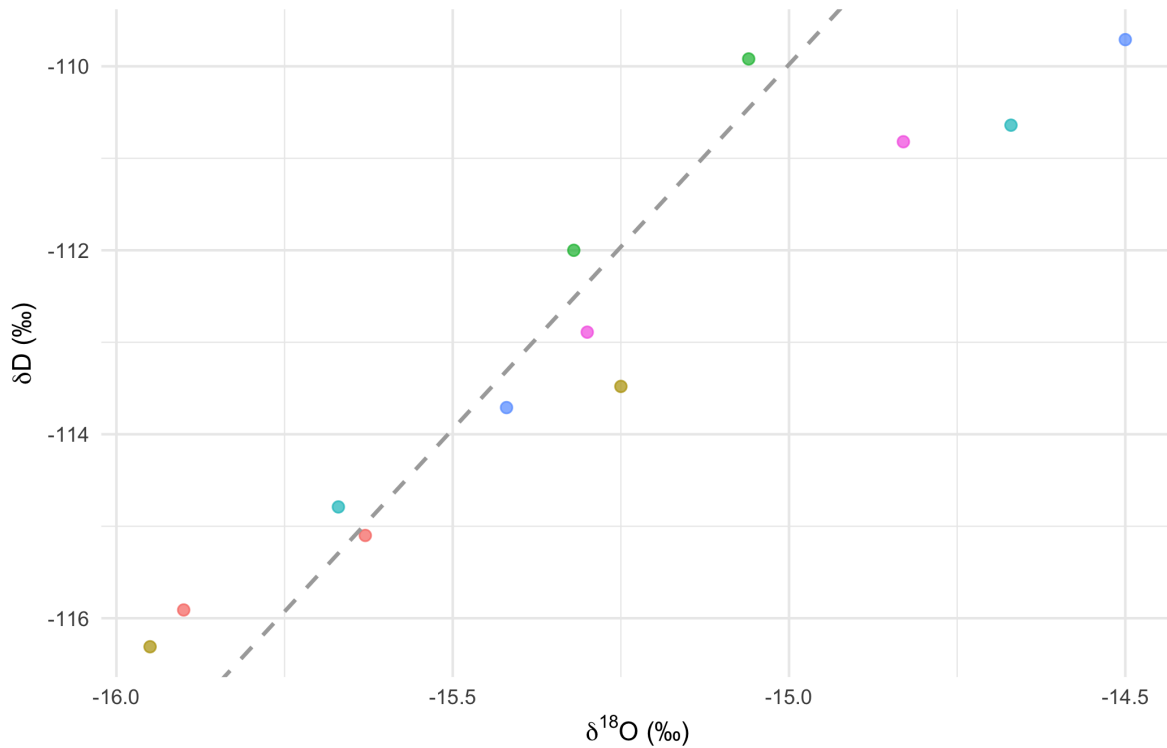


Figure 5:

```
# Coal Creek LMWL calculated from OPIC  $y = 7.93x + 8.97$ 
plot5_1 <- ggplot(coal_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 8.97, slope = 7.93, linetype = "dashed", color = "darkgray", linewidth = 1)
labs(
  title = "Dual-Isotope Plot:  $\delta^{18}O$  vs  $\delta^2H$ ",
  subtitle = "Dashed line = LMWL",
  x = expression(paste(delta^{18}, "O (\u2030)")),
  y = expression(paste(delta, "D (\u2030)")),
  color = "Site"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_1
```

Dual-Isotope Plot: $\delta^{18}\text{O}$ vs δD

Dashed line = LMWL

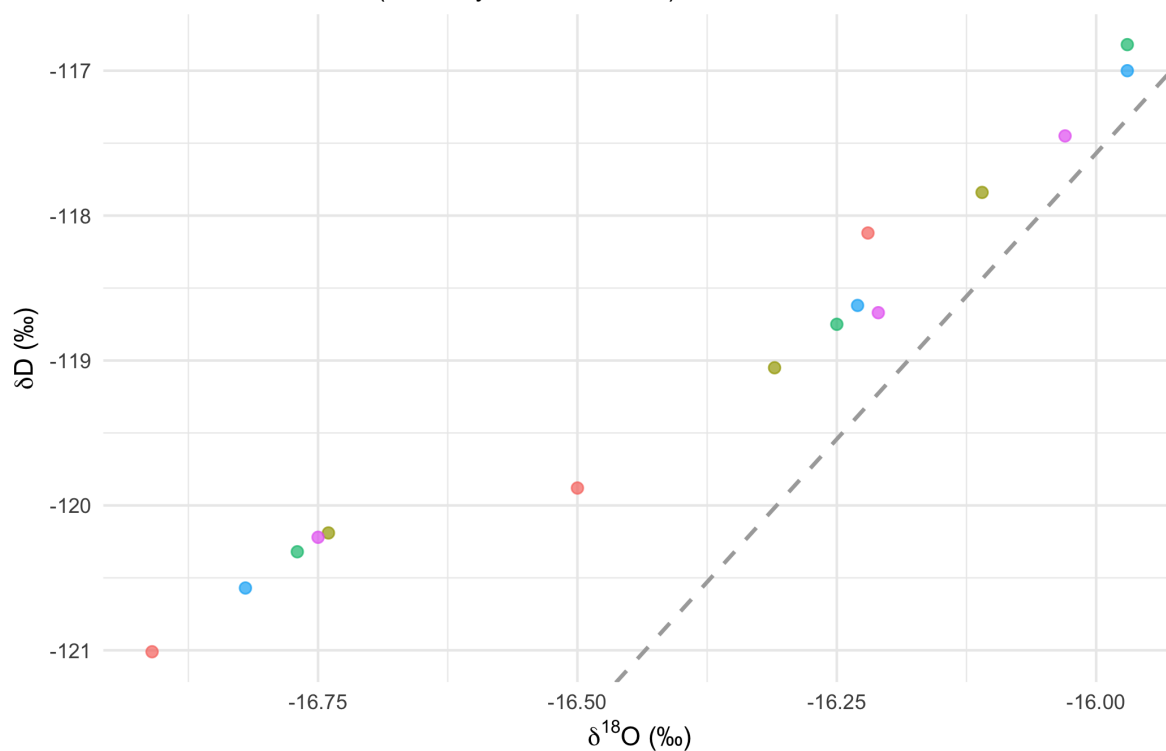


```
# Italian Creek LMWL calculated from OPIC  $y = 7.89 x + 8.67$ 

plot5_2 <- ggplot(italian_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 8.67, slope = 7.89, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.89 x + 8.67$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_2
```

Dual-Isotope Plot: $\delta^{18}\text{O}$ vs δD

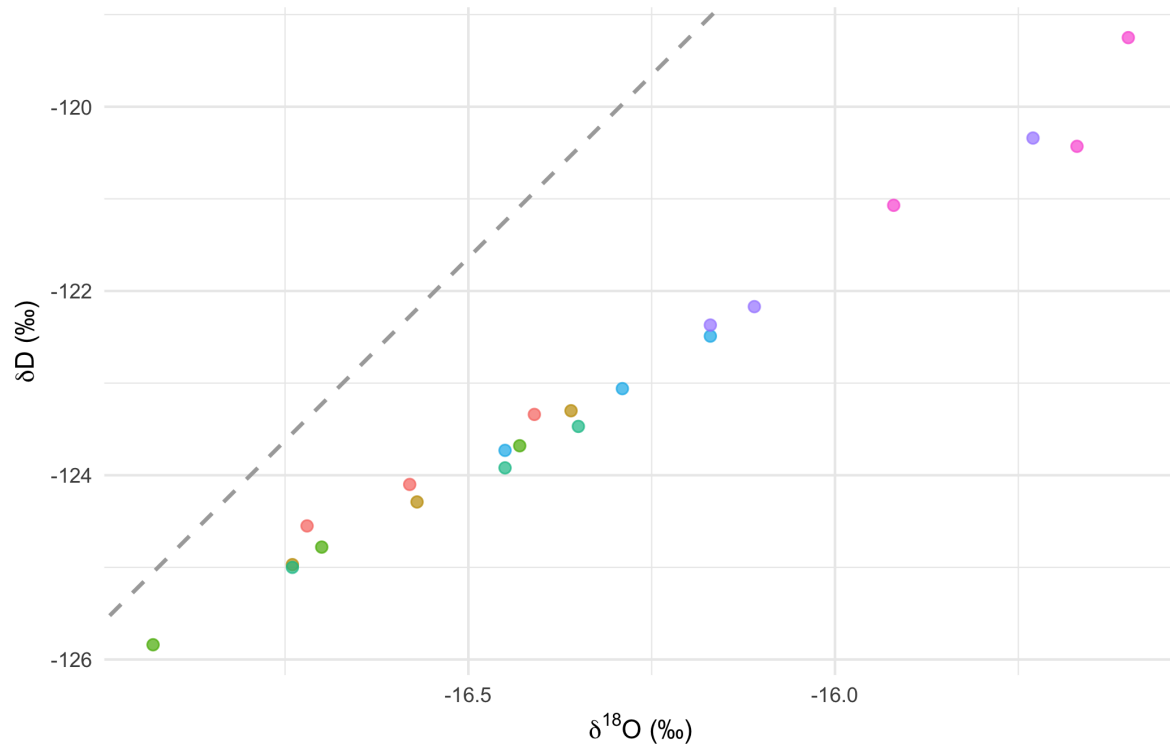
Local Mean Water Line (LMWL: $y = 7.89x + 8.67$)



```
# Trail Creek River Reach LMWL calculated from OPIC  $y = 7.94x + 9.37$ 
plot5_3 <- ggplot(trail_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1)
labs(
  title = "Dual-Isotope Plot with LMWL",
  subtitle = "Local Mean Water Line (LMWL:  $y = 7.94x + 9.37$ )",
  x = expression(paste(delta^{18}, "O (‰)")),
  y = expression(paste(delta, "D (‰)")),
  color = "Site"
) +
theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_3
```

Dual-Isotope Plot with LMWL

Local Mean Water Line (LMWL: $y = 7.94x + 9.37$)



```
figure_5.0 <- plot5_1 + plot5_2 + plot5_3  
figure_5.0
```

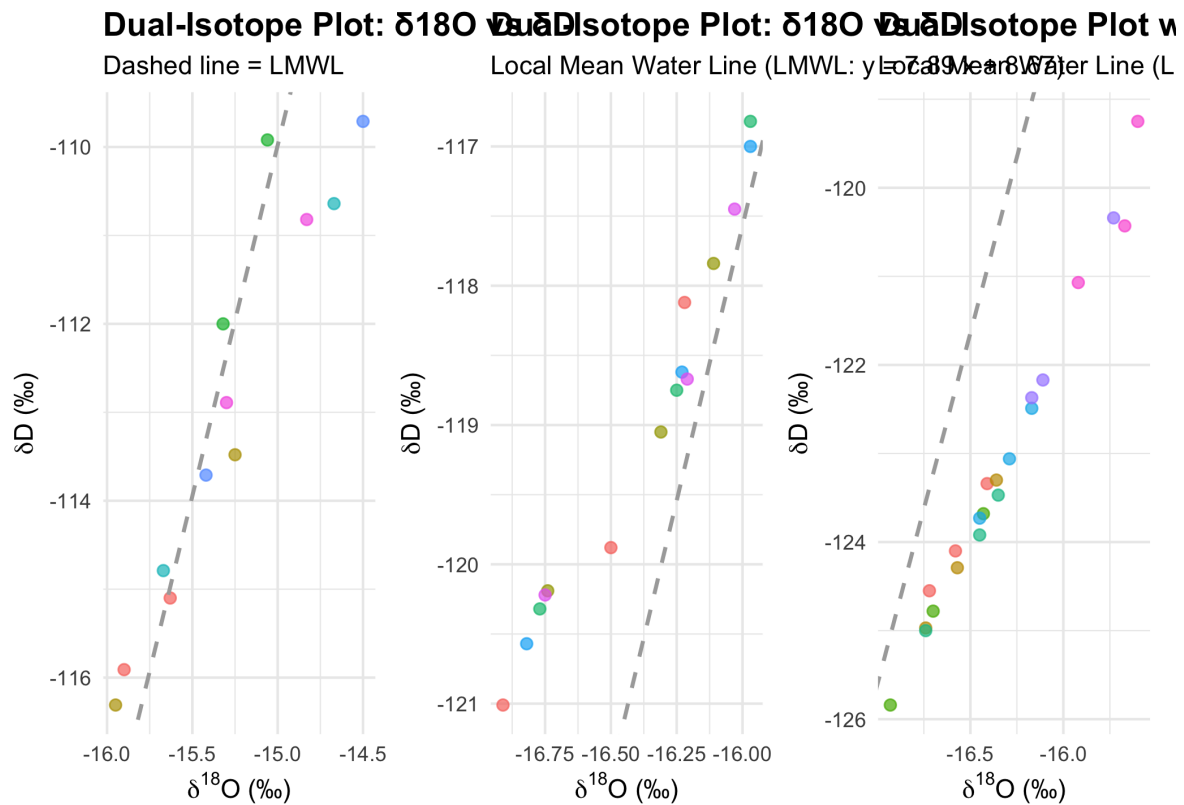
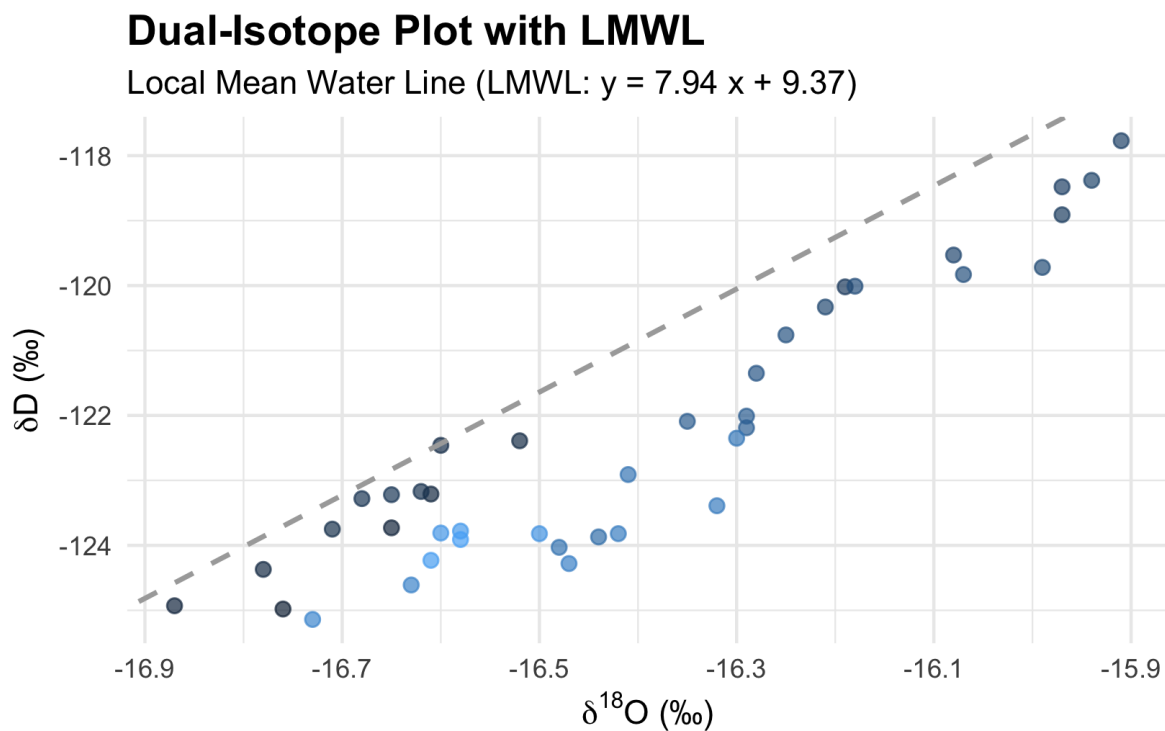



Figure 5.1:

```
# TC ISCO LMWL calculated from OPIC  $y = 7.94 x + 9.37$ 
plot5_4 <- ggplot(tc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 2) +
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL:  $y = 7.94 x + 9.37$ )",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```

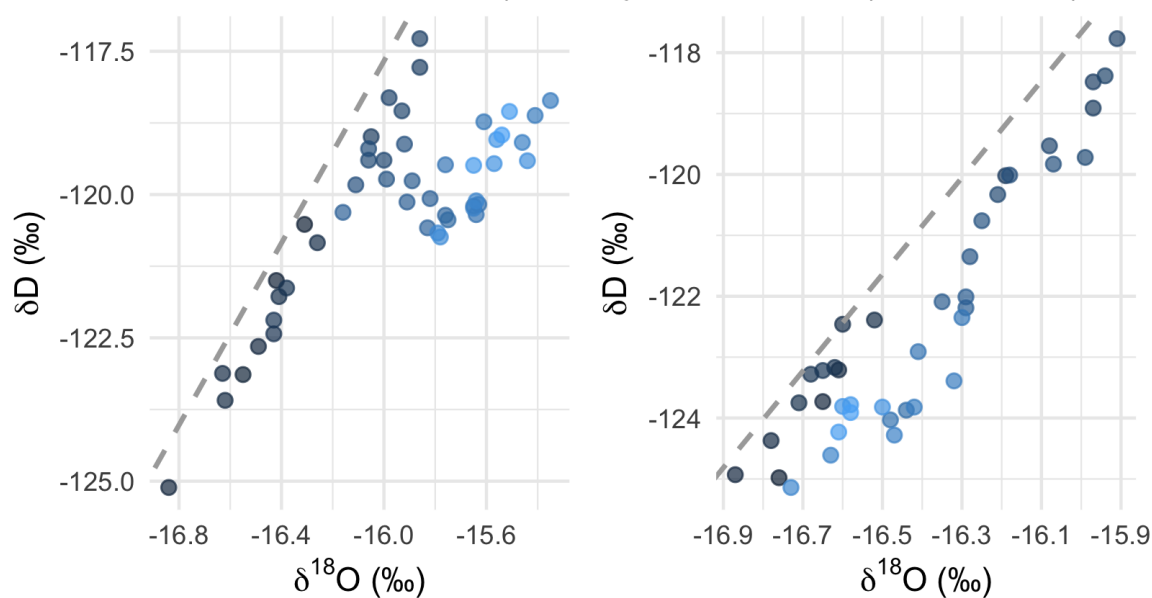
```
# UTC ISCO LMWL calculated from OPIC  $y = 7.94 x + 9.37$ 
plot5_5 <- ggplot(utc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1)
labs(
  title = "Dual-Isotope Plot with LMWL",
  subtitle = "Local Mean Water Line (LMWL:  $y = 7.94 x + 9.37$ )",
  x = expression(paste(delta^{18}, "O (‰)")),
  y = expression(paste(delta, "D (‰)")),
  color = "Site"
) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
plot5_5
```



```
figure_5.1 <- plot5_4 + plot5_5
figure_5.1
```

Dual-Isotope Plot with LMWL

Local Mean Water Line (LMWL: $y = 7.94x + 9.37$)



TC Discharge

```
model_discharge_tc <- lm(lc_excess ~ discharge, data = discharge_tc_isco)
summary(model_discharge_tc)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.13810	-0.33678	-0.07501	0.29166	1.00884

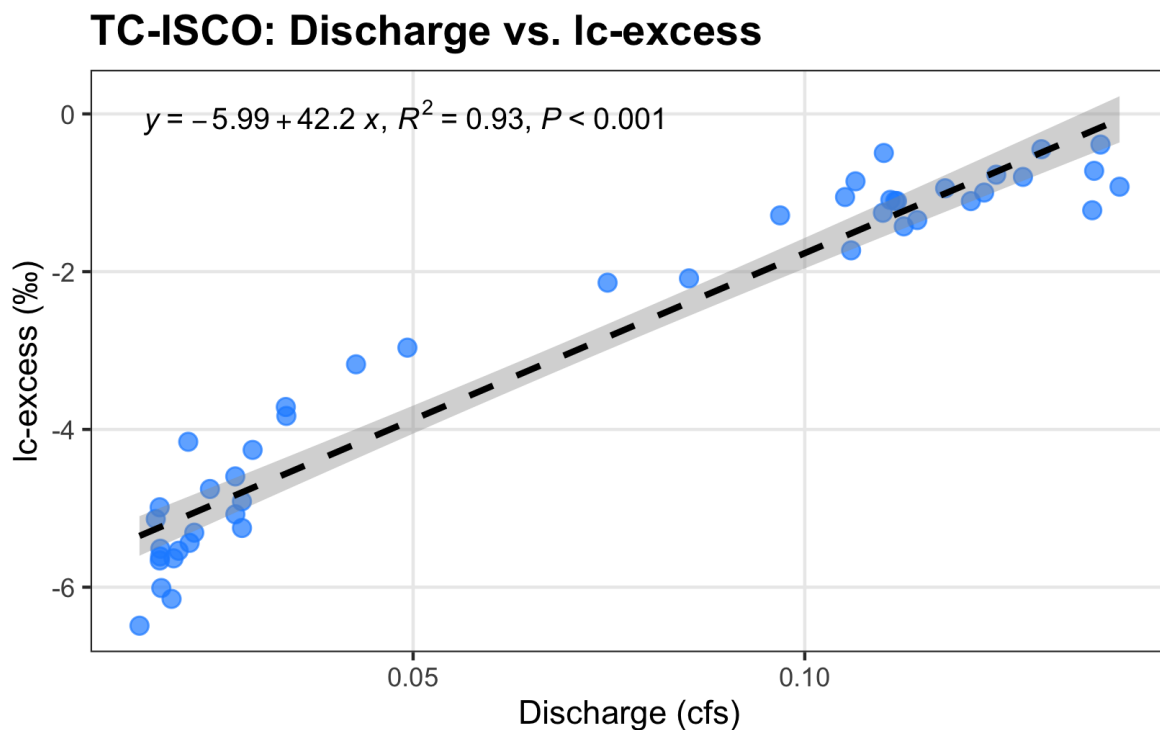
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.986	0.145	-41.28	<2e-16 ***
discharge	42.209	1.719	24.55	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5488 on 44 degrees of freedom
Multiple R-squared: 0.932, Adjusted R-squared: 0.9304
F-statistic: 602.9 on 1 and 44 DF, p-value: < 2.2e-16

```
tc_discharge <- ggplot(discharge_tc_isco, aes(x = discharge, y = lc_excess)) +  
  geom_point(size = 2.5, alpha = 0.7, color = "dodgerblue") +  
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "dashed") +  
  stat_poly_eq(  
    use_label(c("eq", "R2", "P")),  
    formula = y ~ x,  
    label.x = "left", label.y = "top",  
    size = 3.5  
  ) +  
  labs(title = "TC-ISCO: Discharge vs. lc-excess",  
       x = expression(paste("Discharge (cfs)")),  
       y = "lc-excess (\u2030)") +  
  theme_bw() +  
  theme(plot.title = element_text(face = "bold", size = 14),  
        panel.grid.minor = element_blank())  
tc_discharge
```



UTC Discharge

```
model_discharge_utc <- lm(lc_excess ~ discharge, data = discharge_utc_isco)
summary(model_discharge_utc)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_utc_isco)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.39653	-0.58466	0.07884	0.42896	1.38042

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.9751	0.3199	-9.299	2.47e-11 ***
discharge	19.5974	4.1363	4.738	3.00e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.755 on 38 degrees of freedom

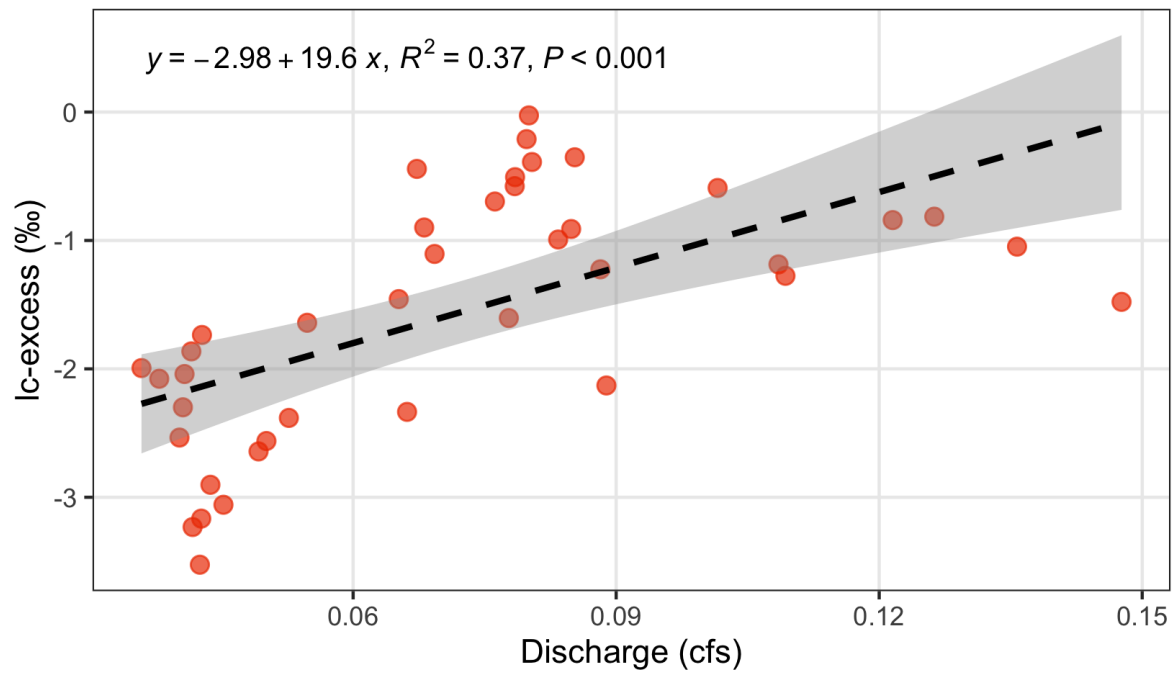
Multiple R-squared: 0.3714, Adjusted R-squared: 0.3548

F-statistic: 22.45 on 1 and 38 DF, p-value: 3e-05

```
utc_discharge <- ggplot(discharge_utc_isco, aes(x = discharge, y = lc_excess)) +
  geom_point(size = 2.5, alpha = 0.7, color = "orangered2") +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "dashed") +
  stat_poly_eq(
    use_label(c("eq", "R2", "P")),
    formula = y ~ x,
    label.x = "left", label.y = "top",
    size = 3.5
  ) +
  labs(title = "UTC-ISCO: Discharge vs. lc-excess",
       x = expression(paste("Discharge (cfs)")),
       y = "lc-excess (\u2030)") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank())

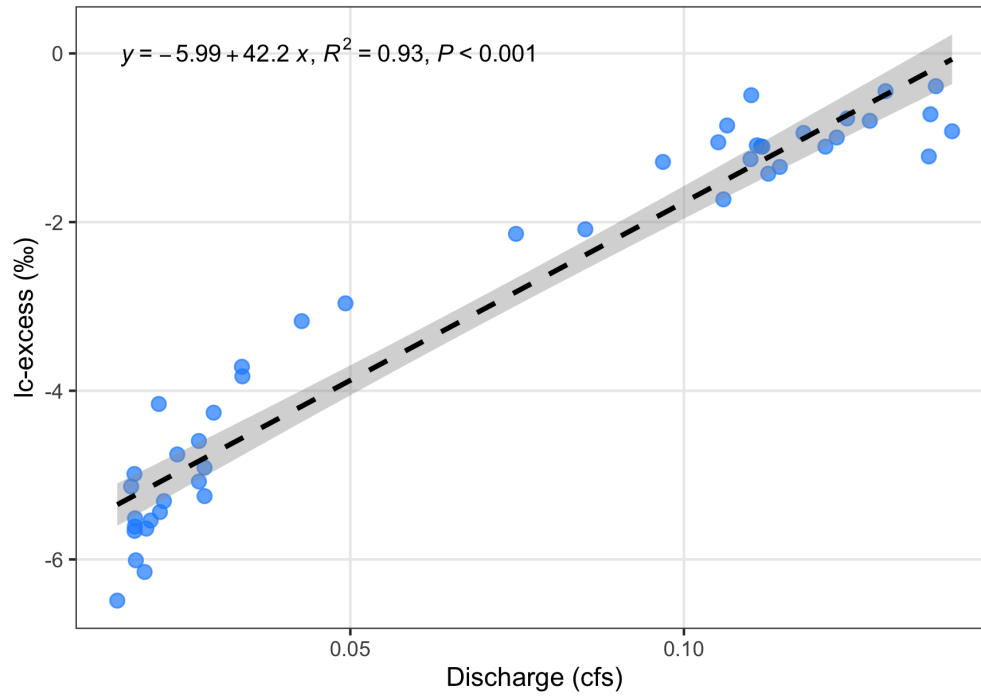
utc_discharge
```

UTC-ISCO: Discharge vs. Ic-excess

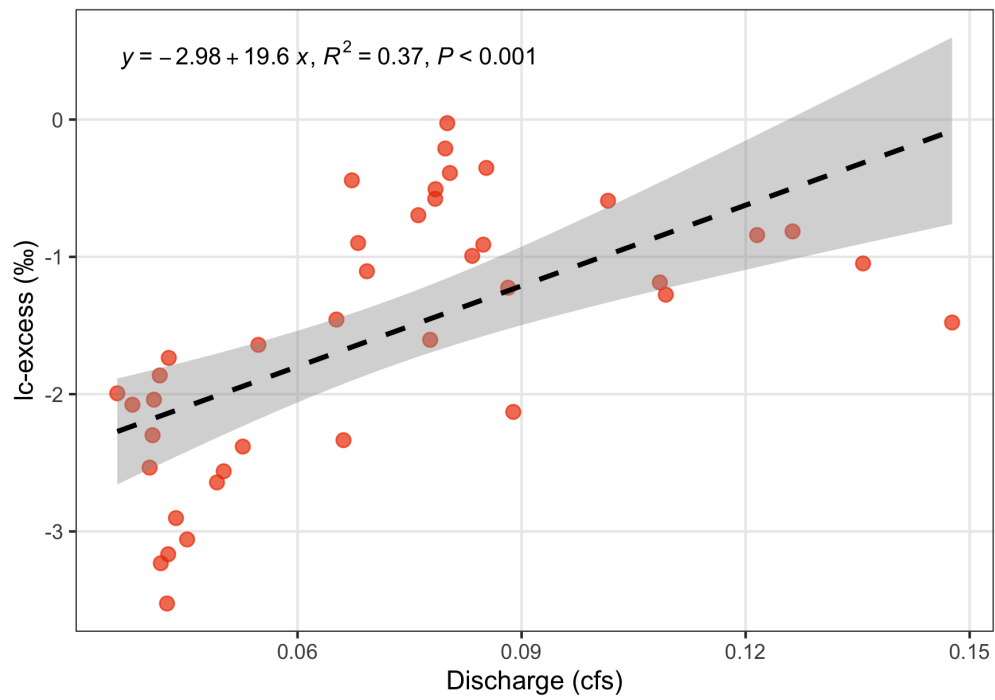


```
combined_stacked_4 <- tc_discharge / utc_discharge  
combined_stacked_4
```

TC-ISCO: Discharge vs. Ic-excess



UTC-ISCO: Discharge vs. Ic-excess



```

ggplot(discharge_isco, aes(x = discharge, y = lc_excess,
                           color = isco,
                           group = isco)) +
geom_point(size = 2.5, alpha = 0.7) +
  scale_color_manual(
    values = c("tc" = "dodgerblue", "utc" = "orangered2"),
    labels = c("tc" = "Trail Creek", "utc" = "Upper Trail Creek")
  ) +
geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black") +
labs(title = "ISCO: Discharge vs. lc-excess",
      x = expression(paste("Discharge (cfs)")),
      y = "lc-excess (\u2030)") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "bottom")

```


ISCO: Discharge vs. Ic-excess

